

Newton's Laws of Motion

(Newton's Laws to Translational Equilibrium)

70 HARD-level MCQs covering all 18 topics of the syllabus

With answer key & solutions

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1. Newton's First Law of Motion

Question 1

HARD

Newton's first law of motion is essentially a statement of which fundamental property of matter?

A. Momentum

B. Inertia ✓

C. Force

D. Impulse

Solution: Newton's first law describes the tendency of a body to resist changes in its state of rest or uniform motion — this tendency is called inertia.

Question 2

HARD

Assertion (A): A passenger standing in a bus tends to fall backward when the bus suddenly starts moving forward.

Reason (R): Due to inertia of rest, the lower part of the passenger's body (in contact with the floor) moves forward with the bus, while the upper part tends to remain at rest.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the mismatch between the sudden forward motion of the feet/floor and the inertia-driven tendency of the upper body to stay at rest causes the backward fall.

Question 3

HARD

Which of the following observations CANNOT be explained using Newton's first law of motion alone?

A. A book remains at rest on a table unless a force is applied.

B. A ball rolling on a frictionless surface continues moving at constant velocity.

C. The acceleration produced in a body is directly proportional to the net force applied. ✓

D. A passenger jerks forward when a moving bus suddenly stops.

Solution: Option C describes the quantitative $F = ma$ relationship, which is the content of Newton's SECOND law, not the first (which addresses only the qualitative tendency to resist changes in motion).

2. Force

Question 4

HARD

Which of the following is a CONTACT force (as opposed to an action-at-a-distance/field force)?

A. Gravitational force

B. Electrostatic force

C. Normal reaction force ✓

D. Magnetic force

Solution: Normal reaction arises only when two surfaces are in physical contact, unlike gravitational, electrostatic and magnetic forces, which act at a distance without contact.

Question 5

HARD

Assertion (A): Force is a vector quantity.

Reason (R): Force has both magnitude and direction and follows the laws of vector addition (e.g., triangle/parallelogram law).

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; possessing both magnitude and direction, and combining via vector addition rules, are exactly what qualifies force as a vector quantity.

3. Inertia

Question 6

HARD

The inertia of a body is a direct measure of its:

A. Weight

B. Mass ✓

C. Volume

D. Velocity

Solution: Inertia — the resistance of a body to a change in its state of motion — is directly proportional to its mass; greater mass means greater inertia.

Question 7

HARD

Which of the following situations best illustrates 'inertia of motion' (as opposed to inertia of rest)?

A. Dust particles fly off a carpet when it is beaten with a stick.

B. A passenger in a moving car is jerked forward when the car suddenly brakes. ✓

C. A coin placed on a card over a glass falls into the glass when the card is flicked away.

D. A person standing in a stationary bus falls backward when the bus suddenly starts.

Solution: Inertia of motion is the tendency of a moving body to continue moving; when the car brakes suddenly, the passenger's already-moving body tends to continue forward, causing the jerk. The other options illustrate inertia of REST.

4. Momentum

Question 8

HARD

The linear momentum of a body is defined as the product of its:

A. Mass and velocity ✓

B. Mass and acceleration

C. Force and time

D. Mass and displacement

Solution: Linear momentum $p = mv$, the product of a body's mass and its velocity.

Question 9

HARD

Assertion (A): Two bodies of different masses can have the same momentum.

Reason (R): Momentum depends on both mass and velocity, so a lighter body moving faster can have the same momentum as a heavier body moving slower.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct — since $p = mv$, a suitable combination of mass and velocity can give the same momentum value for bodies of different mass.

Question 10

HARD

A body of mass 2 kg moving with velocity 3 m/s collides with a stationary body and comes to rest, while the second body moves off with velocity 2 m/s. Assuming conservation of momentum, what is the mass of the second body?

A. 1 kg

B. 2 kg

C. 3 kg ✓

D. 4 kg

Solution: By conservation of momentum: $2 \times 3 = m \times 2 + 0 \Rightarrow m = 6/2 = 3 \text{ kg}$.

Question 11

HARD

Which of the following statements about momentum is INCORRECT?

A. Momentum of a body can be zero even if the body is under the influence of a force at that instant.

B. Momentum is directly proportional to velocity, for constant mass.

C. Momentum is a scalar quantity with no directional aspect. ✓

D. The rate of change of momentum equals the net applied force.

Solution: Momentum is a VECTOR quantity, with direction along the velocity of the body — making this statement incorrect.

5. Newton's Second Law of Motion

Question 12

HARD

According to Newton's second law, the net force acting on a body of constant mass is given by:

A. $F = m/a$

B. $F = ma$ ✓

C. $F = m + a$

D. $F = a/m$

Solution: Newton's second law states $F = ma$ for a body of constant mass, relating net force to the product of mass and acceleration.

Question 13

HARD

A force of 20 N acts on a body of mass 5 kg, initially at rest. What is the velocity of the body after 4 seconds?

A. 8 m/s

B. 16 m/s ✓

C. 4 m/s

D. 20 m/s

Solution: $a = F/m = 20/5 = 4 \text{ m/s}^2$. $v = u + at = 0 + 4 \times 4 = 16 \text{ m/s}$.

Question 14

HARD

Assertion (A): Newton's second law can be expressed as $F = dp/dt$, where p is momentum.

Reason (R): For a system of constant mass, $F = dp/dt$ simplifies to the familiar $F = ma$.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the more general form $F = dp/dt$ reduces to $F = ma$ when mass is constant, since $dp/dt = d(mv)/dt = m(dv/dt) = ma$.

Question 15

HARD

A body of mass 10 kg has its velocity changed from 5 m/s to 15 m/s in 5 seconds under a constant force. What is the magnitude of this force?

- A. 10 N
- B. 20 N ✓**
- C. 30 N
- D. 50 N

Solution: $a = (15-5)/5 = 2 \text{ m/s}^2$. $F = ma = 10 \times 2 = 20 \text{ N}$.

Question 16

HARD

Which of the following statements about Newton's second law is INCORRECT?

- A. It provides a quantitative measure of force.
- B. Acceleration produced is directly proportional to the applied force, for constant mass.
- C. Acceleration produced is directly proportional to the mass of the body, for constant force. ✓**
- D. It reduces to Newton's first law when the net force on a body is zero.

Solution: For a constant force, acceleration is INVERSELY proportional to mass ($a = F/m$), not directly proportional — making this statement incorrect.

Question 17

HARD

A variable force $F = 6t$ (in newtons, t in seconds) acts on a body of mass 3 kg, initially at rest. What is the velocity of the body at $t = 2 \text{ s}$?

- A. 2 m/s
- B. 4 m/s ✓**
- C. 6 m/s
- D. 8 m/s

Solution: $F = m(dv/dt) \Rightarrow dv = (6t/3)dt = 2t \cdot dt$. Integrating from 0 to 2: $v = \int_0^2 2t \cdot dt = [t^2]_0^2 = 4 \text{ m/s}$.

6. Impulse

Question 18

HARD

Impulse of a force is defined as:

- A. The product of force and mass
- B. The product of force and the time for which it acts ✓**
- C. The rate of change of force with time
- D. The product of mass and acceleration

Solution: Impulse $J = F \times \Delta t$ (for constant force), or more generally, the time integral of force — and it equals the change in momentum produced.

Question 19

HARD

A ball of mass 0.5 kg moving with velocity 10 m/s strikes a wall and rebounds with the same speed. What is the magnitude of impulse imparted to the ball by the wall?

- A. 0 kg·m/s
- B. 5 kg·m/s
- C. 10 kg·m/s ✓**
- D. 15 kg·m/s

Solution: Impulse = change in momentum = $m(v_f - v_i) = 0.5 \times (-10 - 10) = -10$ kg·m/s; magnitude = 10 kg·m/s.

Question 20

HARD

Assertion (A): A cricketer moves their hands backward while catching a fast-moving ball.

Reason (R): This increases the time of impact, thereby reducing the average force experienced by the hands, as per the impulse-momentum relationship.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; since impulse (= change in momentum) is fixed, increasing the contact time reduces the average force required, exactly as the reason describes.

Question 21

HARD

Which of the following correctly represents the relationship between impulse and momentum?

- A. Impulse equals the initial momentum only.
- B. Impulse equals the change in momentum of the body. ✓**
- C. Impulse is unrelated to momentum.
- D. Impulse equals momentum divided by time.

Solution: The impulse-momentum theorem states that the impulse of a force equals the resulting change in the body's momentum.

7. Rocket Propulsion

Question 22

HARD

Rocket propulsion is based fundamentally on which principle?

- A. Newton's first law of motion
- B. Conservation of linear momentum (Newton's third law) ✓**
- C. Conservation of angular momentum
- D. Archimedes' principle

Solution: A rocket accelerates forward as a reaction to the high-speed ejection of exhaust gases backward — a direct application of momentum conservation and Newton's third law.

Question 23

HARD

The thrust force on a rocket is given by which expression, where dm/dt is the rate of fuel ejection and v_{rel} is the exhaust velocity relative to the rocket?

- A. $F = v_{rel} \times (dm/dt)$ ✓**
- B. $F = m \times v_{rel}$
- C. $F = (dm/dt) / v_{rel}$
- D. $F = m / v_{rel}$

Solution: Rocket thrust is given by $F = v_{rel} \times (dm/dt)$, the product of the relative exhaust velocity and the rate at which mass (fuel) is ejected.

Question 24

HARD

Assertion (A): A rocket can accelerate even in the vacuum of space, where there is no air to push against.

Reason (R): The rocket's forward thrust arises from the reaction to high-speed ejection of exhaust gases, as per Newton's third law, and does not require any external medium.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; rocket thrust depends only on the momentum change of the ejected exhaust gases, requiring no surrounding medium to 'push against'.

8. Newton's Third Law of Motion

Question 25

HARD

Newton's third law of motion states that:

A. Every action has an equal and opposite reaction, and these forces act on the SAME body.

B. Every action has an equal and opposite reaction, and these forces act on DIFFERENT bodies. ✓

C. Force is directly proportional to acceleration.

D. A body at rest remains at rest unless acted upon by an external force.

Solution: Newton's third law states that action and reaction forces are equal in magnitude, opposite in direction, and act on two DIFFERENT interacting bodies — hence they never cancel each other out.

Question 26

HARD

Which of the following statements about action-reaction pairs is INCORRECT?

- A. Action and reaction forces are always equal in magnitude.
- B. Action and reaction forces act in opposite directions.
- C. Action and reaction forces act on the same body and hence cancel each other out. ✓**
- D. Action and reaction forces occur simultaneously.

Solution: Action and reaction forces act on DIFFERENT bodies, not the same body, and therefore do NOT cancel each other out — making this statement incorrect.

Question 27

HARD

Assertion (A): A book resting on a table remains in equilibrium because the weight of the book and the normal reaction from the table are equal and opposite.

Reason (R): The weight of the book and the normal reaction from the table form a Newton's third law action-reaction pair.

- A. Both A and R are true and R is the correct explanation of A.
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false. ✓**
- D. A is false but R is true.

Solution: The assertion is correct (the book is in equilibrium as these two forces balance), but the reason is FALSE — both forces act on the SAME body (the book), so they cannot be an action-reaction pair; true action-reaction pairs always act on two different bodies.

Question 28

HARD

A swimmer pushes water backward with their hands and moves forward as a result. This is best explained by:

- A. Newton's first law
- B. Newton's second law
- C. Newton's third law ✓**
- D. The law of conservation of energy

Solution: The swimmer's backward push on the water (action) results in an equal and opposite forward reaction force from the water on the swimmer, propelling them forward — a direct application of Newton's third law.

9. Free Body Diagram

Question 29

HARD

A free body diagram (FBD) is primarily used to:

- A. Calculate the total mass of a system of connected bodies
- B. Represent all the forces acting on a single isolated body, in order to analyse its motion ✓**
- C. Determine the coefficient of friction between two surfaces
- D. Calculate the momentum of a body before and after a collision

Solution: An FBD isolates a single body and shows every external force acting on it, providing the basis for applying Newton's laws to analyse its motion.

Question 30

HARD

In the free body diagram of a block resting on a horizontal, frictionless surface with no other forces applied, which forces must be shown?

- A. Weight (downward) and normal reaction (upward) only ✓**
- B. Weight, normal reaction, and an applied horizontal force
- C. Only the weight of the block
- D. Only the normal reaction force

Solution: With no applied force and no friction, only two forces act on the block — its weight (downward) and the normal reaction from the surface (upward), which balance each other.

Question 31

HARD

While drawing the FBD of a block placed on a frictionless inclined plane, the weight of the block is typically resolved into components:

- A. Horizontal and vertical
- B. Along the incline (parallel) and perpendicular to the incline ✓**
- C. Only along the incline
- D. Weight cannot be resolved into components

Solution: For inclined plane problems, weight is conveniently resolved into a component along the incline (causing sliding) and a component perpendicular to the incline (balanced by the normal reaction).

Question 32

HARD

Assertion (A): In a free body diagram, internal forces between parts of the same connected system are NOT shown when analysing a single isolated body.

Reason (R): A free body diagram depicts only the external forces acting ON the isolated body from its surroundings or other bodies.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; an FBD isolates the body of interest and shows only the external forces acting on it, excluding any internal forces within the body itself.

10. Normal Reaction

Question 33

HARD

The normal reaction force between two surfaces in contact acts:

A. Parallel to the surfaces in contact

B. Perpendicular to the surfaces in contact ✓

C. At 45° to the surfaces in contact

D. In the direction of motion only

Solution: By definition, the normal reaction force always acts perpendicular to the plane of contact between two surfaces.

Question 34

HARD

A block of mass 5 kg rests on a horizontal surface. An additional vertical downward force of 10 N is applied on the block. What is the normal reaction force exerted by the surface on the block? (Take $g = 10 \text{ m/s}^2$)

A. 40 N

B. 50 N

C. 60 N ✓

D. 70 N

Solution: $N = mg + F = (5 \times 10) + 10 = 50 + 10 = 60 \text{ N}$.

Question 35

HARD

A block of mass 4 kg rests on a horizontal surface. If an upward vertical force is applied such that the normal reaction becomes zero, this indicates that:

A. The block is accelerating downward.

B. The applied upward force exactly balances (or exceeds) the block's weight, so it is on the verge of (or has already) lifted off the surface. ✓

C. The block is in stable equilibrium on the surface.

D. The mass of the block has increased.

Solution: Normal reaction drops to zero only when the net downward force (weight) is exactly balanced or exceeded by the applied upward force, meaning the block is about to lose (or has lost) contact with the surface.

Question 36

HARD

In a lift accelerating upward with acceleration 'a', the normal reaction (apparent weight) experienced by a person of mass 'm' standing inside is given by:

A. $N = mg$

B. $N = mg + ma = m(g+a)$ ✓

C. $N = mg - ma = m(g-a)$

D. $N = ma$ only

Solution: For a person accelerating upward with the lift, applying Newton's second law: $N - mg = ma \Rightarrow N = m(g+a)$, giving an apparent weight greater than mg .

11. Motion of Bodies in Contact

Question 37

HARD

Two blocks of masses 2 kg and 3 kg are placed in contact on a frictionless horizontal surface. A horizontal force of 20 N is applied to the 2 kg block, pushing both blocks together. What is the common acceleration of the system?

A. 2 m/s^2

B. 4 m/s^2 ✓

C. 6 m/s^2

D. 10 m/s^2

Solution: Total mass = $2+3 = 5 \text{ kg}$. $a = F/M = 20/5 = 4 \text{ m/s}^2$.

Question 38

HARD

For the same system (2 kg and 3 kg blocks in contact, 20 N applied on the 2 kg block), what is the contact force between the two blocks?

A. 8 N

B. 10 N

C. 12 N ✓

D. 20 N

Solution: Considering the 3 kg block alone: contact force = $m \times a = 3 \times 4 = 12 \text{ N}$.

Question 39

HARD

Assertion (A): When two blocks in contact are pushed by a force applied on one of them, the contact force between the blocks depends on which block the force is applied to.

Reason (R): The system's common acceleration remains the same regardless of which block the force is applied to, but the internal contact force differs based on which block must be accelerated via the contact push.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; while the overall acceleration is fixed by total mass and applied force, the contact force needed to accelerate the 'downstream' block does change depending on which block the external force is applied to.

Question 40

HARD

Three blocks of masses m_1 , m_2 and m_3 are placed in contact on a frictionless surface, with a force F applied on the first block. Which expression correctly gives the acceleration of the system?

A. $a = F / m_1$

B. $a = F / (m_1 + m_2 + m_3)$ ✓

C. $a = F \times (m_1 + m_2 + m_3)$

D. $a = F / m_3$

Solution: For blocks moving together with the same acceleration, Newton's second law applied to the whole system gives $a = F / (\text{total mass}) = F / (m_1 + m_2 + m_3)$.

12. System of Masses Tied by Strings

Question 41

HARD

Two blocks of masses 3 kg and 2 kg are connected by a light, inextensible string on a frictionless horizontal surface. A horizontal force of 25 N is applied to the 3 kg block, pulling the system. What is the acceleration of the system?

- A. 3 m/s²
- B. 4 m/s²
- C. 5 m/s² ✓
- D. 6 m/s²

Solution: Total mass = 3+2 = 5 kg. $a = F/M = 25/5 = 5 \text{ m/s}^2$.

Question 42

HARD

For the same system (3 kg pulling 2 kg via a string, with 25 N applied), what is the tension in the connecting string?

- A. 5 N
- B. 10 N ✓
- C. 15 N
- D. 20 N

Solution: Considering the 2 kg block alone: $T = m \times a = 2 \times 5 = 10 \text{ N}$.

Question 43

HARD

Assertion (A): In a system of blocks connected by a light, inextensible string over a frictionless pulley, the magnitude of acceleration is the same for all connected blocks.

Reason (R): Since the string is inextensible, all points along it move with the same speed, and hence the same acceleration.

- A. Both A and R are true and R is the correct explanation of A. ✓
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; the inextensibility constraint directly forces all connected masses to share the same magnitude of speed and acceleration at every instant.

Question 44

HARD

A string can only PULL, never PUSH, on the objects it connects. Which of the following correctly explains why?

A. Strings have negligible mass.

B. A string can only exert tension — a force acting along its length that pulls the connected objects toward each other, never pushes them apart. ✓

C. Strings are always inextensible.

D. Strings are always massless and frictionless over pulleys.

Solution: Tension is fundamentally a 'pulling' internal force generated by a taut string; a slack or compressed string cannot transmit a pushing force, unlike a rigid rod.

Question 45

HARD

Two masses of 5 kg and 3 kg, connected by a light string, hang vertically over a frictionless, massless pulley (Atwood's machine). What is the acceleration of the system? (Take $g = 10 \text{ m/s}^2$)

A. 1.25 m/s^2

B. 2.5 m/s^2 ✓

C. 5 m/s^2

D. 10 m/s^2

Solution: $a = (m_1 - m_2)g / (m_1 + m_2) = (5 - 3) \times 10 / (5 + 3) = 20/8 = 2.5 \text{ m/s}^2$.

13. Pulley Systems

Question 46

HARD

For the Atwood's machine above (5 kg and 3 kg masses connected over a pulley), what is the tension in the connecting string?

A. 30 N

B. 37.5 N ✓

C. 40 N

D. 45 N

Solution: $T = 2m_1m_2g / (m_1 + m_2) = 2 \times 5 \times 3 \times 10 / 8 = 300/8 = 37.5 \text{ N}$.

Question 47

HARD

In an ideal (massless, frictionless) pulley system, the tension throughout a single continuous string is:

- A. Different on either side of the pulley
- B. The same throughout the string ✓**
- C. Zero at the pulley
- D. Dependent on the direction of motion

Solution: For a massless, frictionless pulley, the tension is uniform throughout the length of the string passing over it.

Question 48

HARD

Assertion (A): A massless, frictionless pulley merely changes the direction of the tension force without altering its magnitude.

Reason (R): For a massless pulley, the net torque about its axis must be zero, requiring equal tension on both sides of the pulley.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; a zero net torque requirement (since the pulley is massless, hence has zero moment of inertia) forces the tensions on both sides to be equal in magnitude.

Question 49

HARD

In a movable pulley system where one end of the string is fixed and the pulley itself supports the load, with a force F applied to the free end, what is the ideal mechanical advantage (assuming 2 string segments support the load)?

- A. Mechanical advantage = 1 (no advantage)
- B. Mechanical advantage = 2 ✓**
- C. Mechanical advantage = 0.5
- D. Mechanical advantage = 4

Solution: With two string segments supporting the movable pulley and load, the applied effort is effectively doubled in its lifting effect, giving an ideal mechanical advantage of 2.

Question 50

HARD

A block of mass 4 kg is connected via a string over a frictionless pulley (fixed at the edge of a table) to another block of mass 6 kg; the 4 kg block hangs vertically while the 6 kg block rests on the horizontal frictionless table. What is the acceleration of the system? (Take $g = 10 \text{ m/s}^2$)

A. 2 m/s^2

B. 4 m/s^2 ✓

C. 6 m/s^2

D. 8 m/s^2

Solution: $a = (\text{hanging mass}) \times g / (\text{total mass}) = 4 \times 10 / (4 + 6) = 40 / 10 = 4 \text{ m/s}^2$.

Question 51

HARD

For the system above (4 kg hanging, 6 kg on the table, connected via a pulley), what is the tension in the string?

A. 16 N

B. 20 N

C. 24 N ✓

D. 30 N

Solution: Considering the 6 kg block on the table: $T = m \times a = 6 \times 4 = 24 \text{ N}$.

Question 52

HARD

Which of the following statements about pulley systems is INCORRECT?

A. A fixed pulley changes only the direction of the applied force, not its magnitude.

B. A movable pulley can provide a mechanical advantage greater than 1.

C. In every pulley system, the effort must always be greater than the load being lifted. ✓

D. Combinations of fixed and movable pulleys can be used to achieve larger mechanical advantages.

Solution: With a movable pulley or a compound pulley system, the required effort can actually be LESS than the load, since the mechanical advantage can exceed 1 — making this statement incorrect.

14. Spring Force

Question 53

HARD

According to Hooke's law, the restoring force exerted by a spring is given by:

A. $F = kx$ (in the direction of extension)

B. $F = -kx$ (opposing the direction of extension/compression) ✓

C. $F = kx^2$

D. $F = k/x$

Solution: Hooke's law states $F = -kx$, where the negative sign indicates that the restoring force always opposes the direction of displacement (extension or compression) of the spring.

Question 54

HARD

Assertion (A): The spring constant (k) of a spring is a measure of its stiffness.

Reason (R): A spring with a higher spring constant requires a greater force to produce the same extension/compression, compared to a spring with a lower spring constant.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; since $F = kx$, a higher k directly means more force is needed for the same displacement, which is exactly what 'stiffness' quantifies.

Question 55

HARD

Two identical springs, each with spring constant k , are connected in SERIES. What is the effective spring constant of the combination?

A. $2k$

B. $k/2$ ✓

C. k

D. $4k$

Solution: For springs in series: $1/k_{\text{eff}} = 1/k + 1/k = 2/k \Rightarrow k_{\text{eff}} = k/2$.

Question 56

HARD

Two identical springs, each with spring constant k , are connected in PARALLEL. What is the effective spring constant of the combination?

A. $2k$ ✓

B. $k/2$

C. k

D. $k/4$

Solution: For springs in parallel, the effective spring constant is the simple sum: $k_{\text{eff}} = k + k = 2k$.

15. Frame of Reference

Question 57

HARD

An inertial frame of reference is one in which:

A. Newton's first law of motion holds true (a body remains at rest or in uniform motion unless acted upon by a net external force) ✓

B. Only rotational motion is observed

C. All observers see the same acceleration, regardless of relative motion between frames

D. Pseudo forces must always be introduced to explain motion

Solution: An inertial frame is, by definition, one in which Newton's first law holds without needing any additional (pseudo) forces — i.e., a non-accelerating frame.

Question 58

HARD

Assertion (A): A frame of reference attached to the Earth's surface is, strictly speaking, a non-inertial frame.

Reason (R): The Earth undergoes both rotational motion about its axis and orbital motion around the Sun, both of which involve acceleration.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; Earth's rotational and orbital motions both involve continuous acceleration, technically making an Earth-fixed frame non-inertial (though the effect is negligible for most everyday calculations).

Question 59

HARD

Which of the following is an example of a NON-inertial frame of reference?

A. A car moving at constant velocity on a straight road

B. A train accelerating out of a station ✓

C. A stationary laboratory on Earth's surface (approximately, for most practical purposes)

D. A spacecraft moving at constant velocity in deep space, far from gravitational influence

Solution: An accelerating train is a non-inertial frame, since it is undergoing a change in velocity; the other options describe frames that are either stationary or moving at constant velocity (approximately inertial).

16. Pseudo Force

Question 60

HARD

A pseudo force is introduced in which type of reference frame, to make Newton's laws of motion appear to hold?

A. Inertial frame

B. Non-inertial (accelerating) frame ✓

C. Any frame, inertial or non-inertial

D. Pseudo forces are never needed in any frame

Solution: Pseudo (fictitious) forces are mathematical constructs introduced only in non-inertial (accelerating) frames, to allow Newton's laws to be applied as if the frame were inertial.

Question 61

HARD

The pseudo force experienced by an object of mass m in a frame accelerating with acceleration ' a ' (relative to an inertial frame) is given by:

A. $F_{\text{pseudo}} = ma$, in the SAME direction as the frame's acceleration

B. $F_{\text{pseudo}} = ma$, in the direction OPPOSITE to the frame's acceleration ✓

C. $F_{\text{pseudo}} = m/a$

D. $F_{\text{pseudo}} = a/m$

Solution: The pseudo force has magnitude ma and acts in the direction opposite to the acceleration of the non-inertial frame, in order to correctly account for the object's observed motion within that frame.

Question 62

HARD

Assertion (A): A passenger in a car that suddenly brakes feels thrown forward.

Reason (R): In the non-inertial (decelerating) frame of the car, a pseudo force acts on the passenger in the forward direction (opposite to the car's deceleration), explaining the forward jerk.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; from within the decelerating car's frame, a forward-directed pseudo force correctly accounts for why the passenger appears to be thrown forward.

Question 63

HARD

A block is placed on the floor of a lift that is accelerating upward with acceleration 'a'. In the (non-inertial) reference frame of the lift, the pseudo force on the block acts:

- A. Upward, with magnitude ma
- B. Downward, with magnitude ma ✓**
- C. Horizontally, with magnitude ma
- D. There is no pseudo force in this scenario

Solution: Since the lift accelerates upward, the pseudo force on the block (in the lift's frame) acts downward, with magnitude ma , opposite to the frame's acceleration.

Question 64

HARD

Which of the following statements about pseudo (fictitious) forces is INCORRECT?

- A. Pseudo forces do not arise from any real physical interaction between two bodies.
- B. Pseudo forces obey Newton's third law, having an equal and opposite reaction force acting on another body. ✓**
- C. Pseudo forces are needed only when analysing motion from a non-inertial (accelerating) reference frame.
- D. The magnitude of the pseudo force equals the mass of the object times the acceleration of the non-inertial frame.

Solution: Since pseudo forces are not real physical interactions (they arise purely from the choice of an accelerating frame), they do NOT have a genuine Newton's-third-law reaction force acting on any other body — making this statement incorrect.

Question 65

HARD

In a rotating (non-inertial) frame, the pseudo force experienced by an object, directed radially outward, is known as:

- A. Coriolis force
- B. Centrifugal force ✓**
- C. Centripetal force
- D. Gravitational force

Solution: The centrifugal force is the radially outward pseudo force experienced by an object when analysed from within a rotating (non-inertial) reference frame.

17. Mechanical Advantage

Question 66

HARD

Mechanical advantage of a simple machine (such as a pulley system) is defined as the ratio of:

- A. Input force to output (load) force
- B. Output (load) force to input (effort) force ✓**
- C. Distance moved by effort to distance moved by load
- D. Work input to work output

Solution: Mechanical advantage = load force / effort force, indicating how much the machine amplifies the applied effort.

Question 67

HARD

Assertion (A): An ideal (frictionless, massless) pulley system can provide a mechanical advantage greater than 1.

Reason (R): In such a system, a smaller applied effort can lift a larger load, at the cost of the effort having to move through a proportionally greater distance.

- A. Both A and R are true and R is the correct explanation of A. ✓**
- B. Both A and R are true but R is NOT the correct explanation of A.
- C. A is true but R is false.
- D. A is false but R is true.

Solution: Both statements are correct; by the principle of conservation of energy, any gain in mechanical advantage (force) in an ideal machine is exactly offset by a proportional increase in the distance the effort must travel.

18. Translational Equilibrium

Question 68

HARD

A body is said to be in translational equilibrium when:

A. The net force acting on it is zero, resulting in zero (or constant) linear acceleration. ✓

B. The net torque acting on it is zero, regardless of net force.

C. The body is necessarily at rest.

D. The body is undergoing uniform circular motion.

Solution: Translational equilibrium requires only that the net force on the body be zero; the body may be at rest OR moving at constant velocity — not necessarily at rest.

Question 69

HARD

Three coplanar forces of equal magnitude F act on a point object, each separated from its neighbours by 120° . What is the net force on the object?

A. F

B. $2F$

C. $3F$

D. 0 (Zero) ✓

Solution: For three equal-magnitude forces symmetrically arranged at 120° to each other, the vector sum is exactly zero, leaving the object in translational equilibrium.

Question 70

HARD

Assertion (A): A body can be in translational equilibrium even while multiple forces act upon it simultaneously.

Reason (R): Translational equilibrium requires only that the VECTOR SUM of all forces acting on the body be zero, not that no forces act on it at all.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; equilibrium concerns the net (vector sum) force being zero, which is entirely compatible with several individual forces acting simultaneously, as long as they balance out.